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(54) **RETRACTABLE CURTAIN FOR LIVESTOCK STRUCTURES**

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(52) **U.S. Cl.**

CPC **A01K 1/007** (2013.01); **E06B 9/13** (2013.01); **E06B 9/174** (2013.01); **E06B 9/264** (2013.01); **E06B 9/42** (2013.01); **E06B 9/58** (2013.01); **E06B 9/62** (2013.01); **E06B 9/64** (2013.01); **E06B 9/66** (2013.01); **E06B 9/78** (2013.01); **E06B 2009/2405** (2013.01); **E06B 2009/583** (2013.01)

(58) **Field of Classification Search**

CPC A01K 1/007; E06B 9/264; E06B 9/42; E06B 9/58; E06B 9/66; E06B 9/78; E06B 2009/2405

See application file for complete search history.

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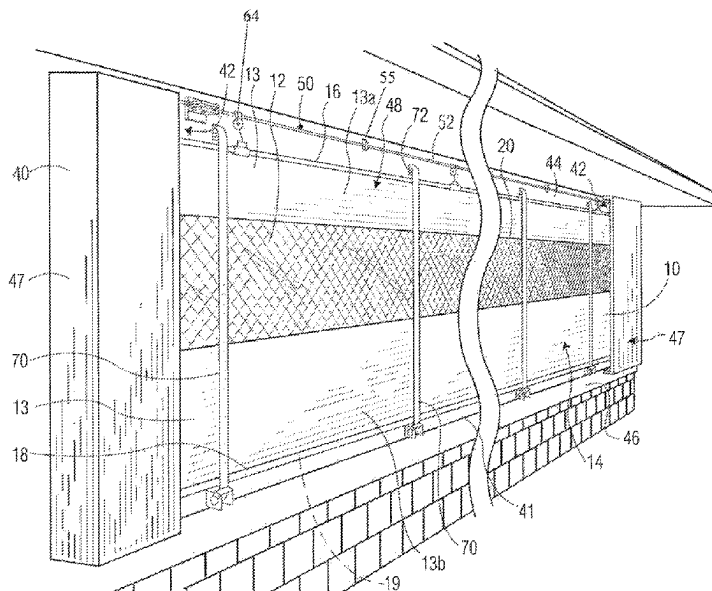
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(57) **ABSTRACT**

A retractable curtain system is provided and includes a curtain support system, a curtain, and a retraction system. The curtain support system includes a pair of outer walls and an upper frame to provide a curtain receiving space. The curtain covers the curtain receiving space and includes a first non-transparent panel and transparent panel connected to the first non-transparent panel. The retraction system is connected to the curtain to lower and raise the transparent panel.

20 Claims, 8 Drawing Sheets



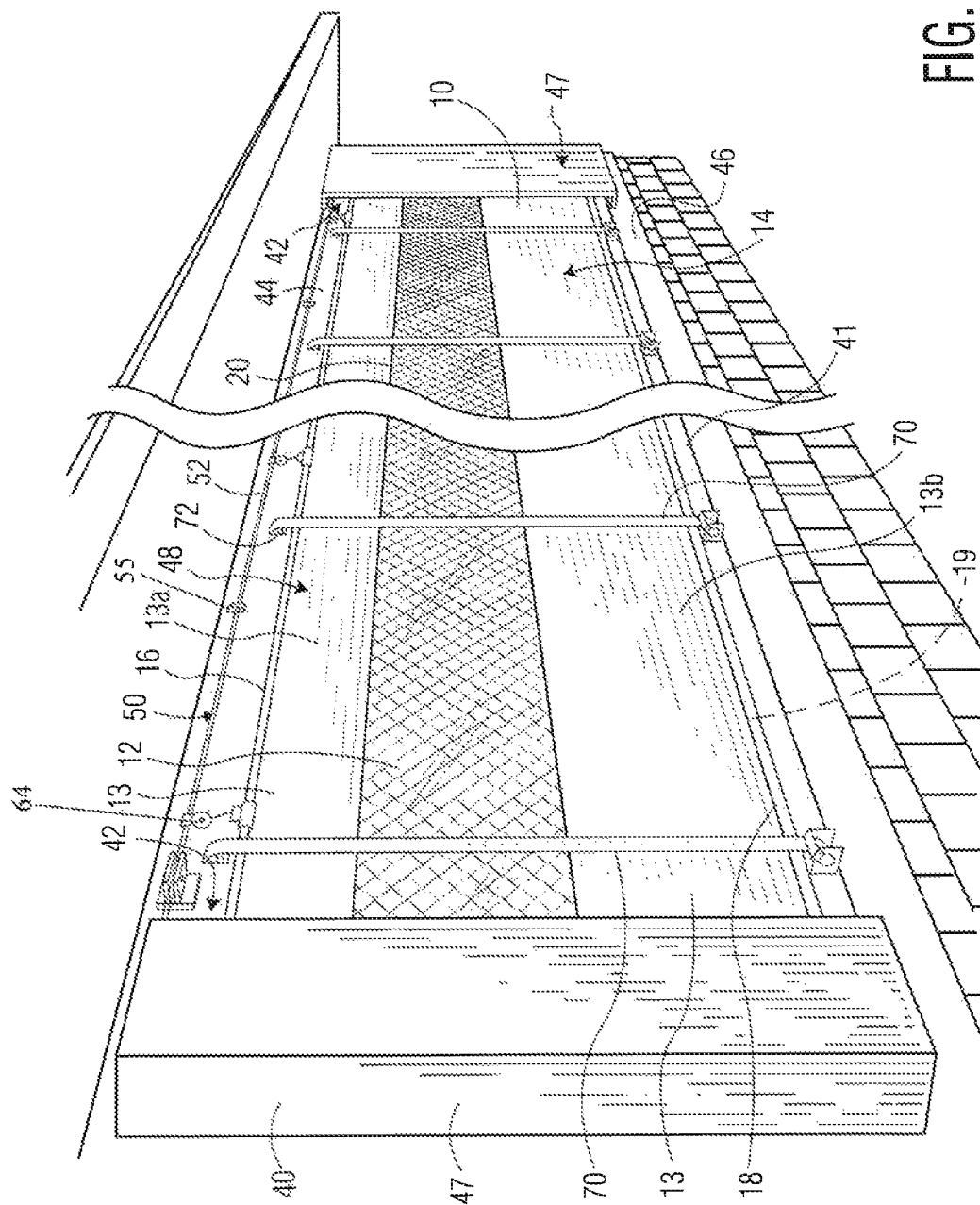
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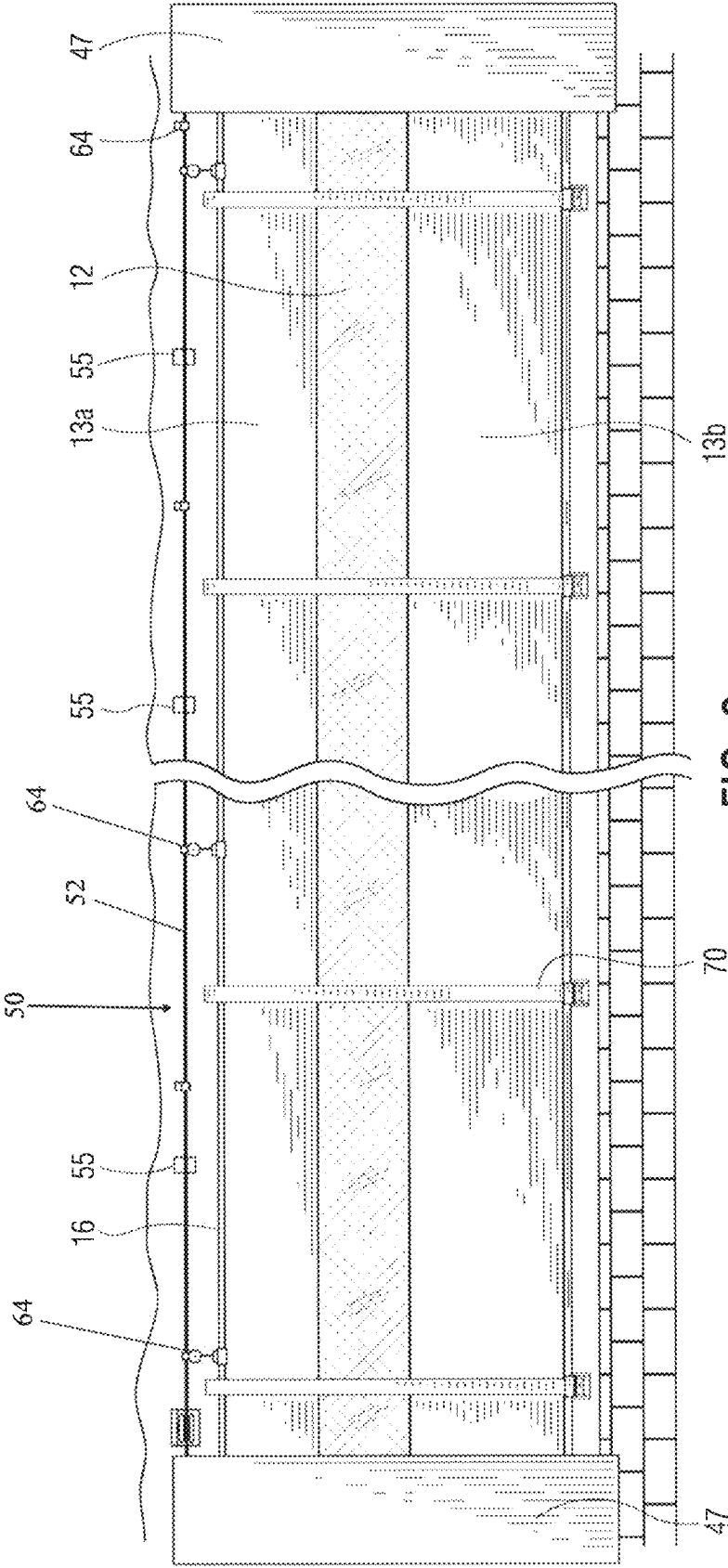


FIG. 2

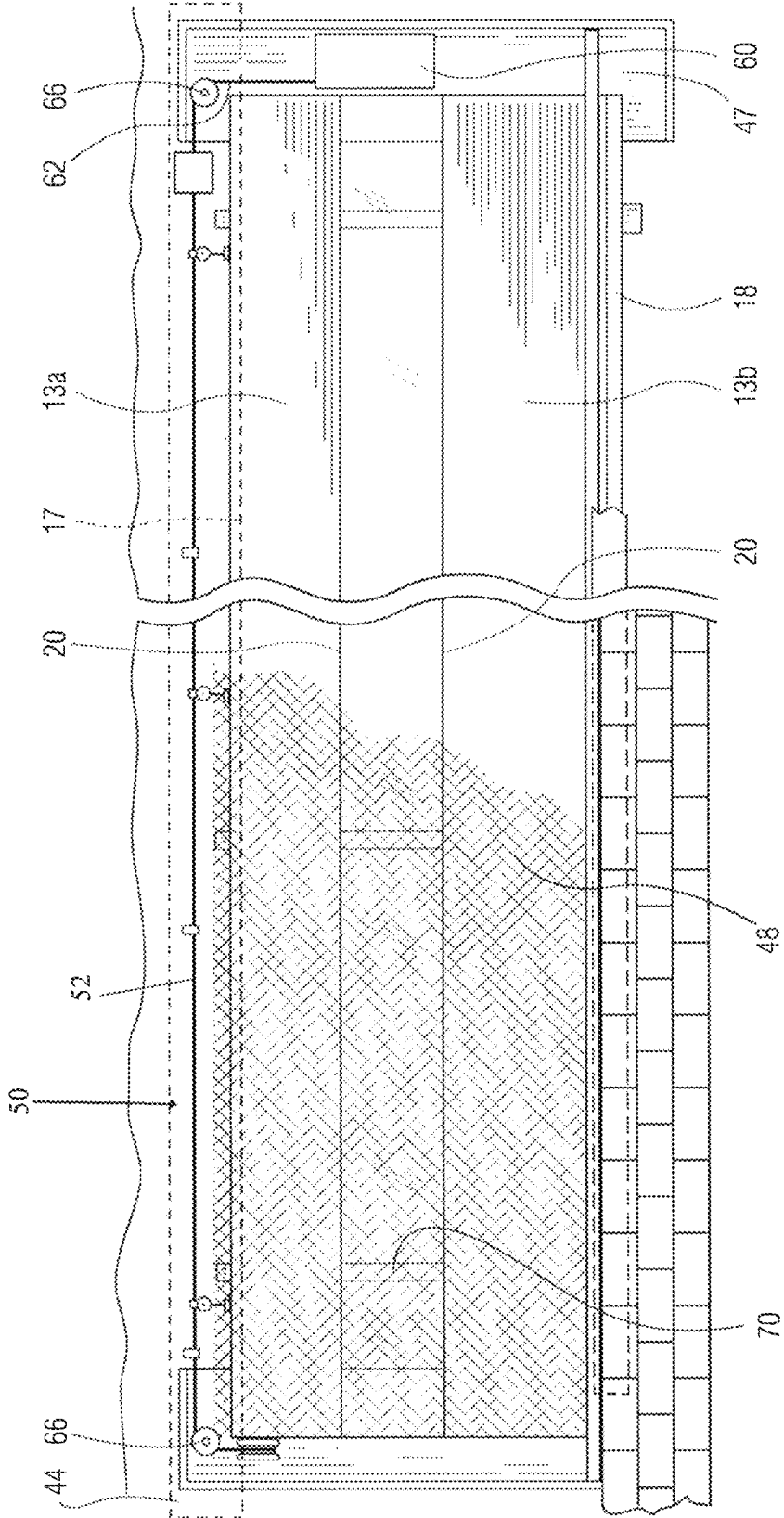
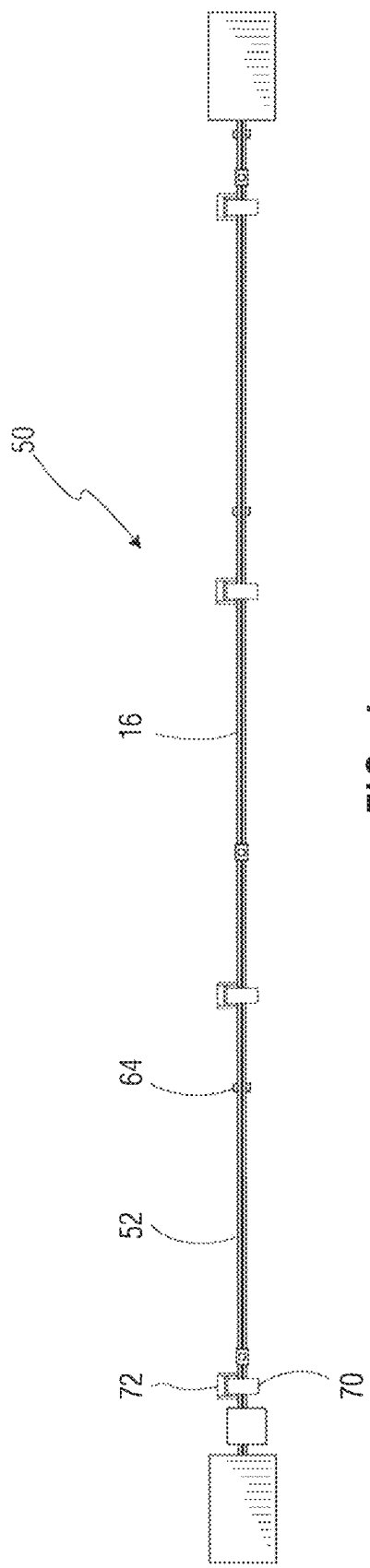


FIG. 3



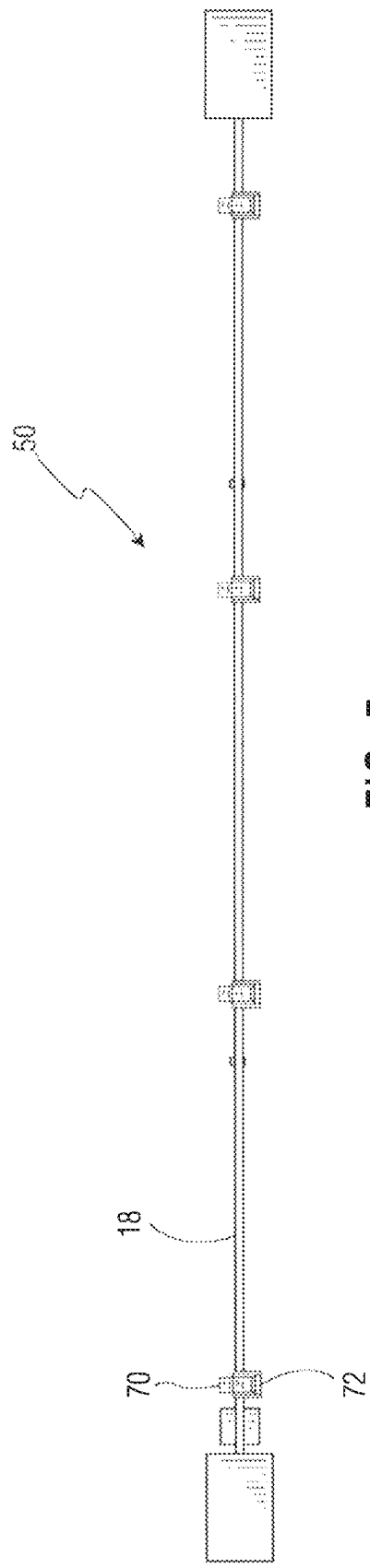


FIG. 5

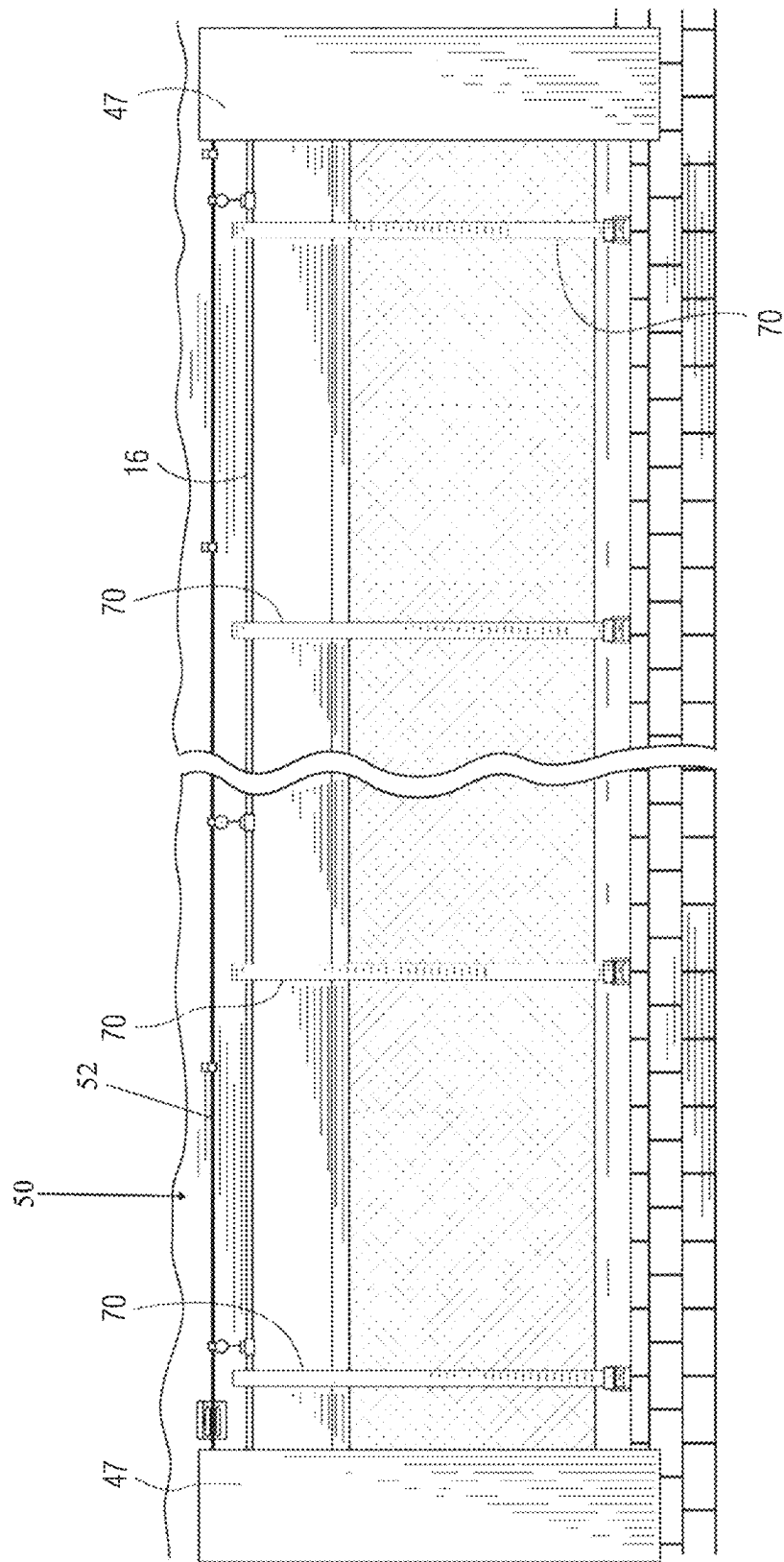


FIG. 6

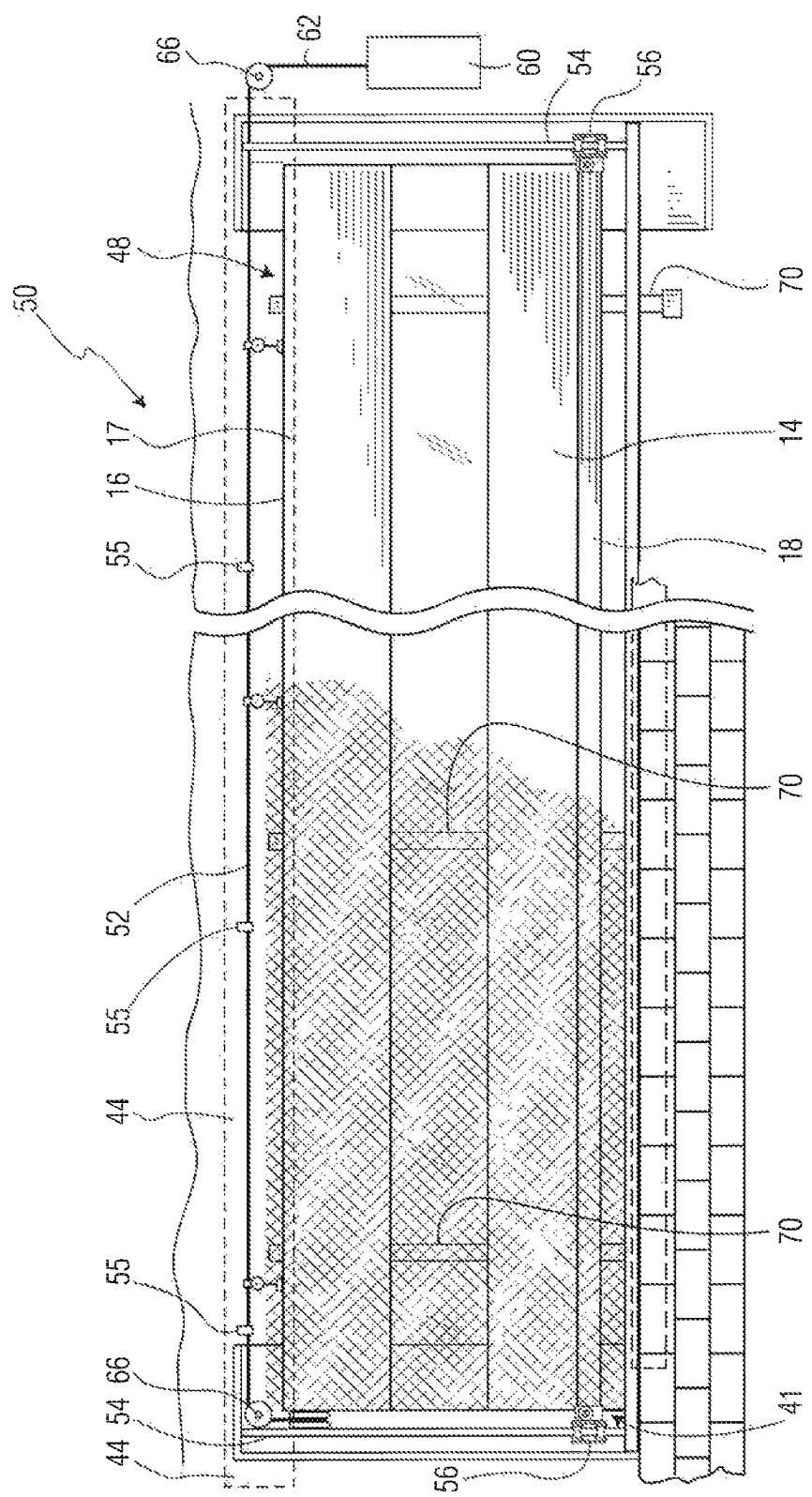


FIG. 7

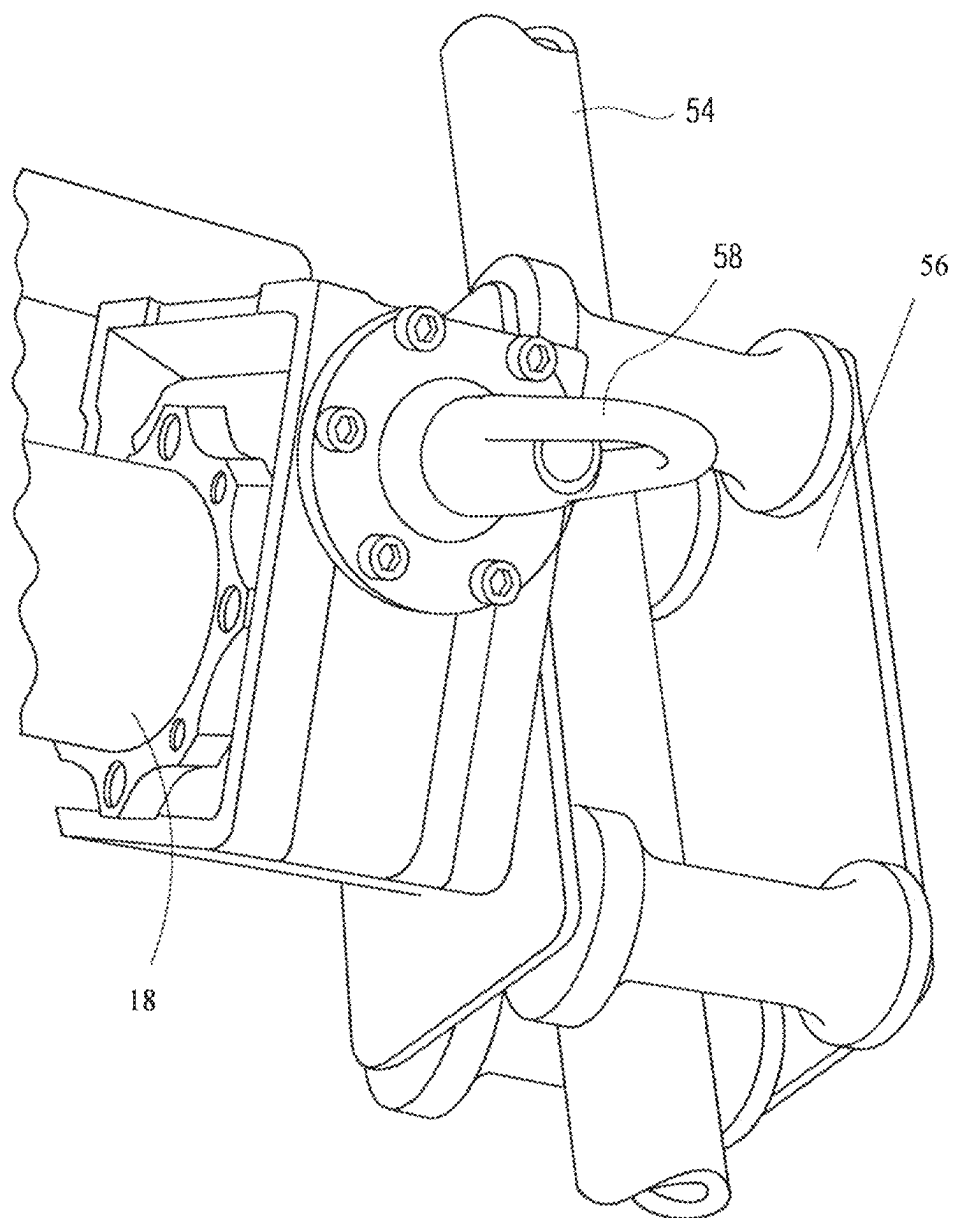


FIG 8

1

RETRACTABLE CURTAIN FOR LIVESTOCK STRUCTURES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. Non-Provisional patent application Ser. No. 17/464,059, filed on Sep. 1, 2021, which is a continuation of U.S. Non-Provisional patent application Ser. No. 14/995,765, filed on Jan. 14, 2016, the contents of which are incorporated herein by reference.

FIELD OF THE INVENTION

The invention relates to a retractable curtain and, more particularly, to a retractable curtain system for livestock structures.

BACKGROUND

Modern livestock management practices typically make use of structures in which to house and shelter livestock for at least a portion of time. The structures offer protection from environmental conditions such as temperature, sun exposure, precipitation, wind, and dust that are harmful or stressful to the livestock. The structures typically possess a roof that covers the structure. Where the structure is constructed with open walls, retractable curtain systems are typically positioned at the open walls. The curtain systems may be extended or not as dictated by environmental conditions. The unrolling or extending of curtain systems, while limiting the entry of undesired environmental factors listed above, have the concurrent effect of limiting the amount of natural light reaching the interior of the structure. The limited light reduces the ability of workers or automated surveillance systems, either from inside or outside of the structure, to visually inspect the livestock and conditions contained within the structure. Furthermore, the limited light also may negatively affect the livestock held in the structure. The limited light may also increase the need for artificial light and thus increase energy costs.

As livestock operations grow, the size of the structures containing livestock may grow in square footage to accommodate larger herds. The increased dimensions of the structure affect the amount and reach of light entering into the interior of the structure. Efficient management of these larger structures is a challenge that requires a curtain system that admits light, limits energy use, and is easy to install, operate, and maintain.

SUMMARY

A retractable curtain system is provided and includes a curtain support system, a curtain, and a retraction system. The curtain support system includes a pair of outer walls and an upper frame to provide a curtain receiving space. The curtain covers the curtain receiving space and includes a first non-transparent panel and a transparent panel connected to the first non-transparent panel. The retraction system is connected to the curtain to lower and raise the transparent panel.

BRIEF DESCRIPTION OF THE DRAWINGS

The invention will now be described by way of example with reference to the accompanying figures, of which:

2

FIG. 1 is a perspective view of a retractable curtain system according to the invention;

FIG. 2 is a front view of the retractable curtain system of FIG. 1;

FIG. 3 is a rear view of the retractable curtain system of FIG. 1;

FIG. 4 is a top view of the retractable curtain system of FIG. 1;

FIG. 5 is a bottom view the retractable curtain system of FIG. 1;

FIG. 6 a rear view of the retractable curtain system of FIG. 1, showing a curtain in a retracted position;

FIG. 7 is a rear view of another embodiment of the retractable curtain assembly according to the invention; and

FIG. 8 is a perspective view of the rolling assembly of the retractable curtain system of FIG. 1.

DETAILED DESCRIPTION OF THE EMBODIMENT(S)

With reference to the accompanying drawings, an embodiment of the invention is directed to a curtain system according to the invention will be described.

With reference to FIGS. 1-6, a retractable curtain assembly 10 according to the invention is shown. The retractable curtain assembly 10 includes the following major components: a retractable curtain 14, a curtain support system 40, and a retraction system 50.

Now with reference to FIGS. 1-4 the retractable curtain assembly 10 will be described. As shown, the retractable curtain assembly 10 includes a curtain 14, an upper curtain support 16, and a lower curtain support 18.

As shown in FIGS. 1-3, the curtain 14 includes at least one or more non-transparent panels 13 and a transparent panel 12 attached to the one or more non-transparent panels 13. The transparent and non-transparent panels 12, 13 are joined or seamed 20 to each other and arranged in generally horizontal and alternating order. One of skill in the art will know appropriate seaming and joining techniques for the materials chosen for the transparent and non-transparent panels 12, 13. In the embodiment shown, the transparent panel 12 and non-transparent panels 13a and 13b are attached to each other in such an order that light is introduced into the structure through the transparent panel 12.

The transparent panel 12 is a sheet of material sufficiently transparent to allow the passage of light. Non-limiting examples of material suitable for the transparent panel 12 include plastic materials known to those of skill in the art. In particular, the transparent panel 12 has a haze less than 4.5% and light transmittance greater than 90%, both when tested according to ASTM 1003-00.

The non-transparent panel 13 is also a sheet of plastic film or fabric material. In particular, in the embodiment shown, the non-transparent panel 13 is opaque. However, one skilled in the art should appreciate that the non-transparent panel 13 could also be a translucent material which transmits light, though at a transmission rate less than that of the transparent panel 12.

Many suitable materials are known and available. The material chosen for the panels 12, 13 may additionally be abrasion-resistant, weatherable, and dimensionally stable. The joined transparent panel 12 and the non-transparent panel 13 are preferably comprised of materials sufficiently strong and flexible to allow the curtain 14 to be rolled into a cylindrical shape without cracking or fracturing as conditions warrant.

3

In the shown embodiment, the curtain **14** includes a pair of non-transparent panels **13a** and **13b** and a transparent panel **12** positioned between the pair of non-transparent panels **13**. More particularly, a first non-transparent panel **13a** is provided and is a rectangular section of material positioned along an upper half of the curtain **14**. A second non-transparent panel **13b** is also provided and is a rectangular section of material that is positioned along a lower half of the curtain **14**. In the embodiment shown, the first non-transparent panel **13a** and the second non-transparent panel **13b** have the same width, but the second non-transparent panel **13b** has greater height than the first non-transparent panel **13a**. The transparent panel **12** is also a rectangular section of material that is positioned between the first and second non-transparent panels **13a**, **13b**. An upper surface of the transparent panel **12** is connected to a lower surface of the first non-transparent panel **13a**, while a lower surface of the transparent panel **12** is connected to an upper surface of the second non-transparent panel **13b**. The transparent panel **12** may be connected to the non-transparent panels **13a**, **13b** using known fastening mechanisms, including stitching, welding, adhesives, etc. One skilled in the art should appreciate that the curtain **14** may be designed differently than shown, and the non-transparent panel **13** and the transparent panel **12** may be arranged in different configurations.

The upper curtain support **16** is rigid and strong enough to support the curtain **14** and the lower curtain support **18**. As a result, the upper curtain support **16** can be any suitable cylindrical tubing, e.g., an aluminum extrusion, or another metal or a reinforced plastic material. Likewise, the lower curtain support **18** can be any suitable cylindrical tubing, e.g., an aluminum extrusion, or another metal or a reinforced plastic material. The lower curtain support **18** can be weighted to maintain the curtain **14** in an upright position, with the weight of the lower curtain support **18**.

One skilled in the art should appreciate that the size and shape of the upper curtain support **16** and the lower curtain support **18** could take other known shapes and be prepared from other known materials.

In the shown embodiment, the curtain **14** further includes an upper support receiving passageway **17** to receive the upper curtain support **16** and a lower support receiving passageway **19** to receive the lower curtain support **18**. The upper support receiving passageway **17** is formed by looping and stitching an upper seam of the curtain **14** to a surface of the curtain **14**. Likewise, the lower support receiving passageway **19** is formed by looping and stitching a lower seam of the curtain **14** to the surface of the curtain **14**. Both the support receiving passageway **17** and the support receiving passageway **19** are tubular passageways in the shown embodiment that are sized to receive the upper curtain support **16** and the lower curtain support **18** respectively.

Now with reference to FIG. 1-4, the curtain support system **40** will be described.

As shown, the curtain support system **40** includes a pair of outer walls **42**, an upper frame **44**, a lower frame **46**, and a curtain receiving space **48**. The pair of outer walls **42** are separately positioned on opposite side from one another with the upper frame **44** and the lower frame **46** extending there between. The curtain receiving space **48** is provided between the pair of outer walls **42**, the upper frame **44**, and the lower frame **46**.

The curtain support system **40** also optionally includes left and right covers **47** which conceal the edges of the curtain receiving space **48**. These covers **47** may be hinged doors that open when necessary.

4

Since livestock structures can be of such a size that the structures may include a plurality of window framings **41** formed along the curtain receiving space **48** between the upper frame **44** and the lower frame **46**. These window framings **41** allow the passage of air to ventilate the structures. In certain embodiments, these window framings **41** are formed in a similar, generally rectilinear shape and are positioned in a side-to-side configuration between opposite ends of the structure. A person of skill in the art will recognize that the structures may be fabricated from a variety of materials and still be in keeping with the scope of the invention. As required by the particular livestock sheltered, the structure may have one or more door or other access that permit the livestock to enter or leave and allow workers to enter and leave to care for and feed the livestock.

Now with reference to FIG. 1-6, the retraction system **50** will be described.

While an embodiment is shown, the retraction system **50** is used to roll and unroll the curtain **14** and any known hoisting system known to the art would be sufficient.

In the shown embodiment, the retraction system **50** includes an upper support rod **52**, and a lifting mechanism **60**.

As shown in the Figures, the upper support rod **52** is a structural member, such as a pole, positioned along and fixed to the upper frame **44**. In the shown embodiment, the upper support rod **52** is tubular and secured to the upper frame **44** using a bracket or similar known device shown, the lifting mechanism **60** is a winch, or the like, such as a windlass, for raising and lowering the curtain **14**. The lifting mechanism **60** in this embodiment is shown as being hand-cranked, but in other embodiments it may be electrically or pneumatically operated. The lifting mechanism **60** includes a plurality of hoisting cables **62** that runs from the lifting mechanism **60** for raising and lowering the curtain **14**. Here there are one or more sheaves **64** attached onto the upper support rod **52**, and the hoisting cables **62** passes through one of each. The cable **62** passes through the inside of the upper support rod **52** and connects to the curtain **14** and, in particular, the upper curtain support **16**. Additionally, pulleys **66** are between the upper support rod **52** and the upper curtain support **16**, and guide the hoisting cable **62**. Here, there is one cable **62** going to the winch through the sheaves **64**.

As shown in FIG. 7, another embodiment of the invention is shown, wherein the retraction system **50** further includes a pair of outer support rods **54**, and a plurality of outer brackets **55**, a pair of rolling mechanisms **56**, and a winding device **58**.

As shown in the Figures, the pair of outer support rods **54** are structural members, such as rigid poles, positioned along and fixed to the pair of outer walls **42**. In the shown embodiment, each outer support rod is a cylindrical pole secured to one of the outer walls using a plurality of outer brackets **55** positioned at opposite ends thereof.

As shown, the pair of rolling mechanisms **56** includes a set of rollers positioned around and corresponding to the pair of outer support rods **54**. Each rolling mechanism **56** is free to move up and down the pair of outer support rods **54**.

As shown, the winding device **58** is connected to one of the rolling mechanisms **56**, along one side of the pair of outer support rods **54**. In the shown embodiment, the rolling mechanisms **56** is a roll-up mechanism, that is, a device for winding up the curtain **14** about the lower curtain support **18**. The lower curtain support **18** is connected to the winding device **58**, so as to rotate when the winding device is rotated, for instance, using the roll-up mechanism.

5

As shown in the FIGS. 1-6, anti-billowing straps 70 are provided and configured as generally shown. The strap 70 itself may be a woven web or other suitable material and is attached at the upper frame 44 by means of a strapping bracket 72, which also serves a stand-off to create a space between the strap 70 and the outward facing side of the upper frame 44. The lower end of the strap 70 is attached in various ways, including attached to the lower frame 46. The curtain 14 can be restrained with the strap 70 preventing billowing inward or outward.

Now with reference to the FIGS. 1-8, operation of the retractable curtain assembly will be discussed. As shown in FIG. 1, the curtain 14 is movable between a retracted position X (see FIG. 4) and an extended position Y (see FIG. 4) to control the amount of light entering the livestock structure. According to the invention, the curtain 14 includes at least one flexible transparent panel 12 connected to at least one flexible non-transparent panel 13.

As shown in FIG. 1-3, when the curtain 14 is extended as fully as possible, light is allowed into the interior of the structure at the same time that there is a barrier to the free flow of wind and other undesirable environmental conditions into and out of the structure. As shown in FIG. 4, the curtain 14 is hoisted into a storage position. The curtain 14 is wound and stored at the general height of the upper framing 44, where the retraction system 50 is attached to the structure. The curtain 14 can be extended or rolled up depending on the weather and the conditions inside the structure. As an option, the curtain 14 may be rolled up partially. A worker may easily crank the curtain 14 using the lifting mechanism 60. The upper curtain support 16 raises the curtain 14 to the upper support rod 52. The worker uses the lifting mechanism 60 to lift or lower the curtain using the hoisting cable 62 that is attached to the upper curtain support 16 and drawn through the pulleys 66. At a change of conditions, it is also a simple matter to wind up the curtain 14 for storage using the retraction system 50.

In addition, as shown in FIG. 6-8, the curtain 14 can then be wound upward from the lower curtain support 18 using the winding device 58. The worker uses the winding device 58 to rotate the lower curtain support to wind up the curtain 14 about the lower curtain support 18. Accordingly, the lower curtain support 18 steadily scales up the outer support rods 54 using the rolling mechanisms 56 that are also attached to the winding device 58 (see FIG. 6).

The foregoing illustrates some of the possibilities for practicing the invention. Many other embodiments are possible within the scope and spirit of the invention. It is, therefore, intended that the foregoing description be regarded as illustrative rather than limiting, and that the scope of the invention is given by the appended claims together with their full range of equivalents.

What is claimed is:

1. A retractable curtain system, comprising:

a curtain support system having a pair of outer walls and an upper frame that defines a curtain receiving space; a curtain capable of covering an entirety of the curtain receiving space and having:

a non-transparent panel connected to an upper curtain support, the upper curtain support at a first end of the curtain and a lower curtain support supported by the curtain and positioned at a second and opposite end of the curtain; and

a retraction system connected to the curtain to lower and raise a transparent panel up and down, the retraction system is configured to roll and unroll the curtain about the lower curtain support.

6

2. The retractable curtain system of claim 1, wherein the transparent panel and the non-transparent panel are attached to each other in such an order that light is introduced into the retractable curtain system through the transparent panel.

3. The retractable curtain system of claim 2, wherein the transparent panel has a haze less than 4.5% and light transmittance greater than 90%.

4. The retractable curtain system of claim 3, wherein the non-transparent panel is opaque.

5. The retractable curtain system of claim 4, wherein the retraction system rolls the curtain into a cylindrical shape.

6. The retractable curtain system of claim 5, wherein an upper surface of the transparent panel is connected to a lower surface of the non-transparent panel.

7. The retractable curtain system of claim 6, wherein the curtain further includes an upper support receiving passageway to receive the upper curtain support and a lower support receiving passageway to receive the lower curtain support.

8. The retractable curtain system of claim 7, wherein the retraction system includes an upper support rod and a lifting mechanism.

9. The retractable curtain system of claim 8, wherein the lower curtain support is connected to a winding device, so as to rotate when the winding device is rotated.

10. The retractable curtain system of claim 9, wherein an anti-billowing strap is attached at the upper frame which create a space between the strap and an outward facing side of the upper frame.

11. A retractable curtain system, comprising:

an upper curtain support;

a lower curtain support; and

a curtain having a non-transparent panel connected to the upper curtain support and a transparent panel extending horizontally and coupled to the non-transparent panel, the upper curtain support is at a first end of the curtain and the lower curtain support is positioned at a second and opposite end of the curtain, the curtain is retracted by being rolled about the lower curtain support.

12. The retractable curtain system of claim 11, wherein the transparent panel is a sheet of material sufficiently transparent to allow the passage of light.

13. The retractable curtain system of claim 12, wherein the non-transparent panel is a sheet of plastic film or fabric material.

14. The retractable curtain system of claim 13, wherein the non-transparent panel is opaque.

15. The retractable curtain system of claim 14, wherein the retractable curtain system further comprises a second non-transparent panel.

16. The retractable curtain system of claim 15, wherein an upper surface of the transparent panel is connected to a lower surface of the non-transparent panel, while a lower surface of the transparent panel is connected to an upper surface of the second non-transparent panel.

17. The retractable curtain system of claim 16, wherein the lower curtain support is weighted to maintain the curtain in an upright position.

18. The retractable curtain system of claim 17, wherein the curtain further includes an upper support receiving passageway to receive the upper curtain support.

19. The retractable curtain system of claim 18, wherein the curtain further includes a lower support receiving passageway to receive the lower curtain support.

20. A retractable curtain system, comprising:

an upper curtain support;

a lower curtain support; and

7

a curtain having a non-transparent panel connected to the upper curtain support and a transparent panel extending horizontally and coupled to the non-transparent panel, the upper curtain support is at a first end of the curtain and the lower curtain support is positioned at a second 5 and opposite end of the curtain, the curtain is retracted by being rolled about the lower curtain support, an unrolled length of the transparent panel is variable between an extended position and a retracted position.

* * * * *

10

8